

Midterm Lab Exam - Submission Form

Student Information System with GitHub Integration

Submission Details

GitHub Repository URL	https://github.com/FundanoJoseph/Midterm-Lab-Exam
Project Folder	student-info-system/
Pull Request	arena/3d68f5bf-midterm-lab-exam -> main
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1. Features Implemented

Student Data Management (30 pts)

- [x] Add new students - validated input, duplicate-ID protection
- [x] View student records - formatted table of all students
- [x] Update student information - partial updates (blank keeps value)
- [x] Delete students - with confirmation prompt
- [x] JSON data storage - data/students.json, atomic save on every change

Cloud-Ready Architecture (25 pts)

- [x] Modular code structure - models/, services/, utils/ layers
- [x] Configuration management - config/config.json + env-var overrides
- [x] Error handling - custom exceptions, corrupt-file recovery, atomic writes
- [x] Logging system - rotating file logger (logs/app.log, 1 MB x 3 backups)

GitHub Integration (25 pts)

- [x] Proper repository structure - src/, data/, config/, logs/, tests/, docs/
- [x] Meaningful commit history - 10+ logical conventional commits
- [x] Branching strategy - feature branch merged to main via pull request
- [x] GitHub features - pull request review + GitHub Actions CI (pytest + smoke test)
- [x] README documentation - setup, configuration, architecture, workflow

Code Quality (20 pts)

- [x] Clean, readable, PEP 8-consistent code
- [x] Docstrings on every module, class and function + inline comments
- [x] Consistent, meaningful naming conventions

Bonus Features (+10 pts)

- [x] Unit tests - 9 pytest tests (CRUD, validation, persistence, search, export, recovery)
- [x] Data validation - ID format, name, age range (5-100), course, email regex
- [x] Search functionality - case-insensitive across ID, name, course, email
- [x] Data export - one-key CSV export to data/exports/
- [x] Advanced error recovery - corrupt JSON backed up and replaced automatically

2. Challenges Faced

1. PEP 668 externally-managed environment - pip install was blocked by the OS. Solved by testing in a virtual

environment and keeping the app 100% standard-library so it runs anywhere with Python 3.8+.

2. Preventing data loss on crash - a plain `json.dump()` can leave a truncated file if the process dies mid-write. Solved with atomic saves: write to a temp file, then `os.replace()`.

3. Recovering from a corrupt data file - a hand-edited `students.json` would crash the app on startup. Solved by catching decode errors, backing the bad file up to `students.json.corrupt`, logging the error, and starting fresh.

4. Keeping the interactive menu clean - console logging interfered with the menu UI. Solved by sending only `ERROR+` to the console; everything goes to the rotating log file.

5. Duplicate log handlers - re-initializing the logger appended duplicate handlers. Solved with a guard in `setup_logger()`.

6. GitHub workflow setup - removing the starter ZIP artifact, keeping `logs/` and `data/exports/` out of version control via `.gitignore`, and structuring commits so the history reads as a clean progression.

3. How to Run

```
git clone https://github.com/FundanoJoseph/Midterm-Lab-Exam.git
cd Midterm-Lab-Exam/student-info-system
python -m src.main          # interactive menu
pytest                      # unit tests (pip install -r requirements.txt)
```

Menu: 1. Add student | 2. View all students | 3. Update student | 4. Delete student | 5. Search students | 6. Export to CSV | 0. Exit